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Studierichting: Informatica
Oefeningenreeks 4A nr. 38.2

$$I = \int \frac{2x+11}{x^2+6x+10}dx = 2\int \frac{x}{x^2+6x+10}dx + 11\int \frac{1}{(x+3)^2+1}dx$$
$$= 2\int \left(\frac{2x+6}{2(x^2+6x+10)} - \frac{3}{x^2+6x+10}\right)dx + 11\int \frac{1}{(x+3)^2+1}dx$$

Stel: $t = x^2+6x+10 \Rightarrow dt = 2x+6dx$ en stel: $y = x+3 \Rightarrow dy = dx$

$$= 2\left[\int \frac{1}{2(x^2+6x+10)} \cdot (2x+6dx) - \int \frac{3}{(x+3)^2+1}dx\right] + 11\int \frac{1}{(x+3)^2+1}dx$$
$$= 2\left[\int \frac{1}{2t}dt - \int \frac{3}{y^2+1}dy\right] + 11\int \frac{1}{y^2+1}dy$$
$$= 2\left[\frac{1}{2}\ln|t| + c - 3\,\mathrm{Bgtan}\,(y) + c\right] + 11\,\mathrm{Bgtan}\,(y) + c$$
$$= \ln|x^2+6x+10| - 6\,\mathrm{Bgtan}\,(x+3) + 11\,\mathrm{Bgtan}\,(x+3) + c$$
$$= \ln|x^2+6x+10| + 5\,\mathrm{Bgtan}\,(x+3) + c$$

References